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- [Pneumonia classifier reaches AUC 0.975, looks at the heart](#)
- [U-Net segments 392 lesions, blames hair](#)
- [Vision-language model writes report for grey square: "benign"](#)
- [Clever Hans \(Berlin, 1904\) still undefeated in arithmetic](#)
- [Audience: coffee supplies "critical", organisers "monitoring"](#)

Sources: team-08, SageMaker, Bedrock



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1.1

Explainability: why did the model say that?

Three questions, three models, one rule:
test every explanation against a control.

team-08 · AI4Imaging Summer School 2026 · Vienna

Three questions, three models, one rule

controls: random weights, a lung mask

Where does it fail, and does it know?

error maps · test-time
augmentation

controls: the worst cases, the
flipped image

Why does it say that, and is the reason real?

reports from Qwen3-VL

controls: another prompt,
another patient, no image

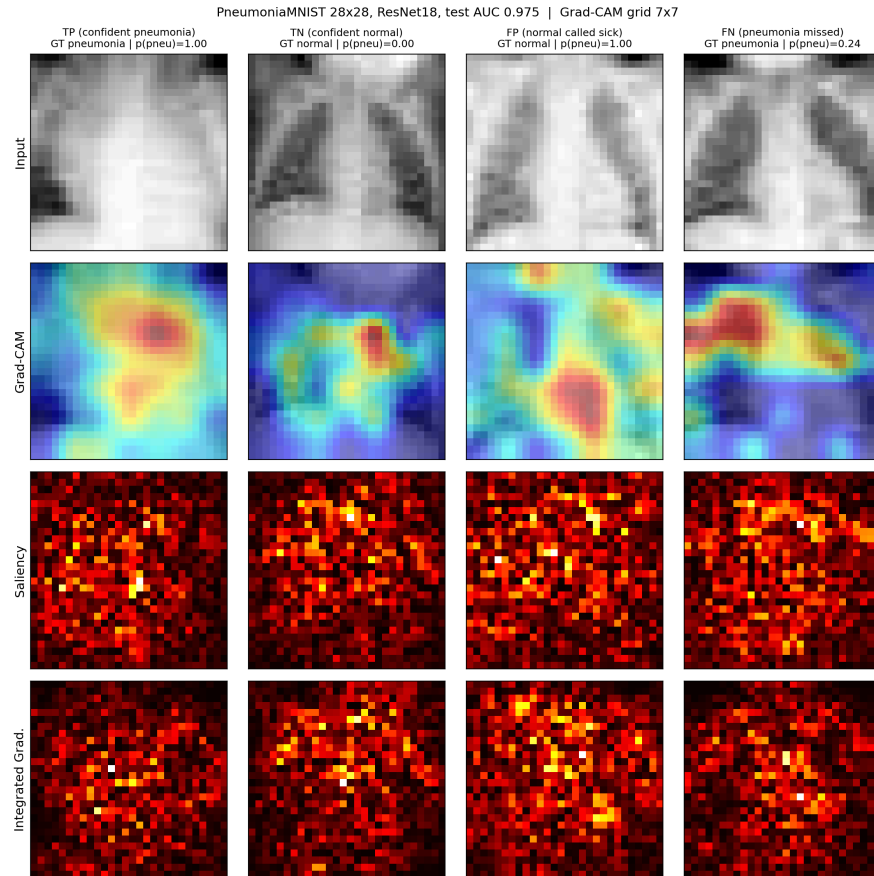


1904: a horse called Clever Hans could do arithmetic. He was reading the questioner's face.

Every model we trained is a potential Clever Hans. The job is finding the face.

Hans & Wilhelm von Osten, c. 1904 (public domain)

Where did it look?



ResNet18 on PneumoniaMNIST, 28×28 pixels. Four cases, three methods: a pneumonia it got right, a normal it got right, a normal it called sick, a pneumonia it missed.

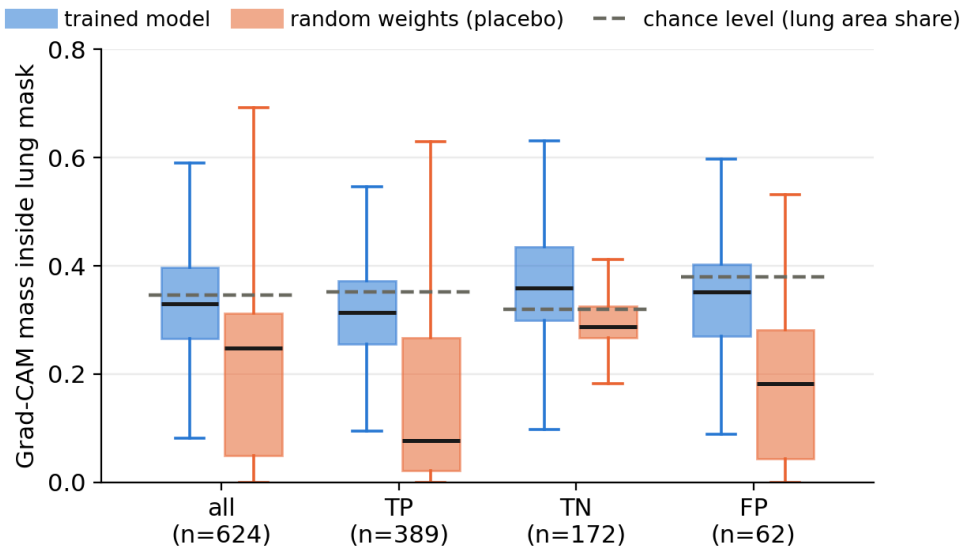
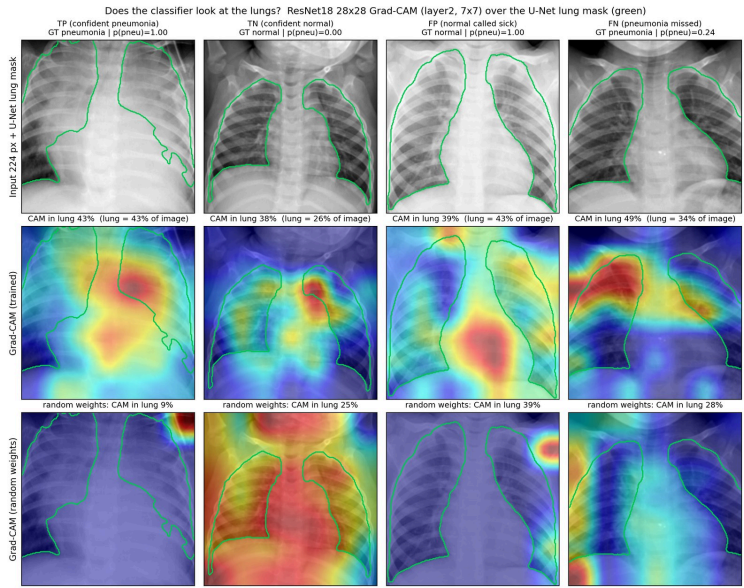
test AUC 0.975

Grad-CAM on **layer2**, grid 7×7, upsampled to 28×28

Saliency and Integrated Gradients: per pixel, sharp but noisy

The heat map is not wrong. It is just not saying much.

Does it look at the lungs?

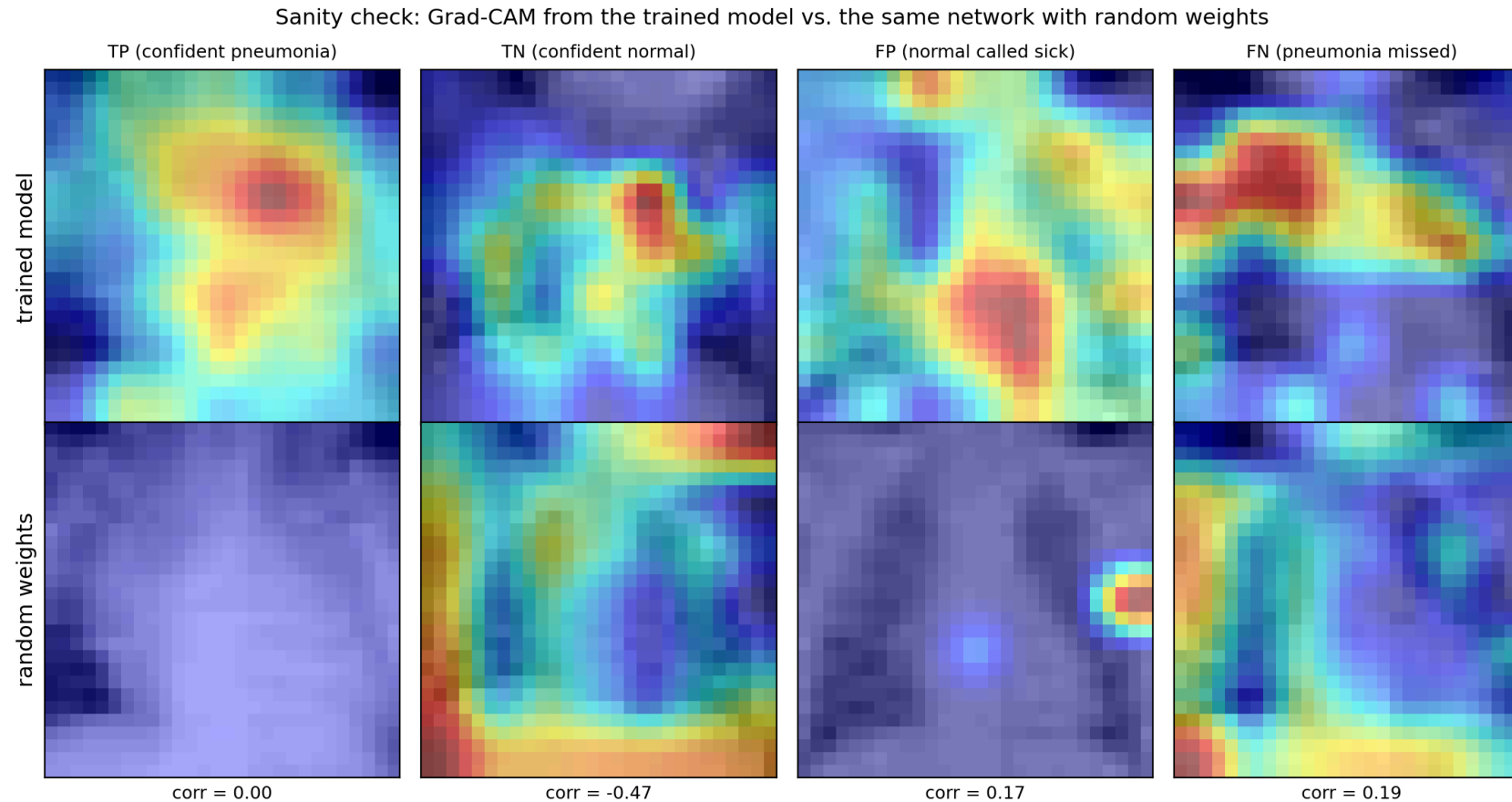


Grad-CAM mass inside the lungs	0.34
lung share of the image, a random gaze	0.35
random weights	0.22
hottest pixel inside the lungs	23% of 624 images

Where are the lungs? A second model says so: a U-Net (ResNet34 encoder) trained on 138 public chest X-rays with hand-drawn lung masks (Montgomery, val Dice 0.98). Green: its lung contour.

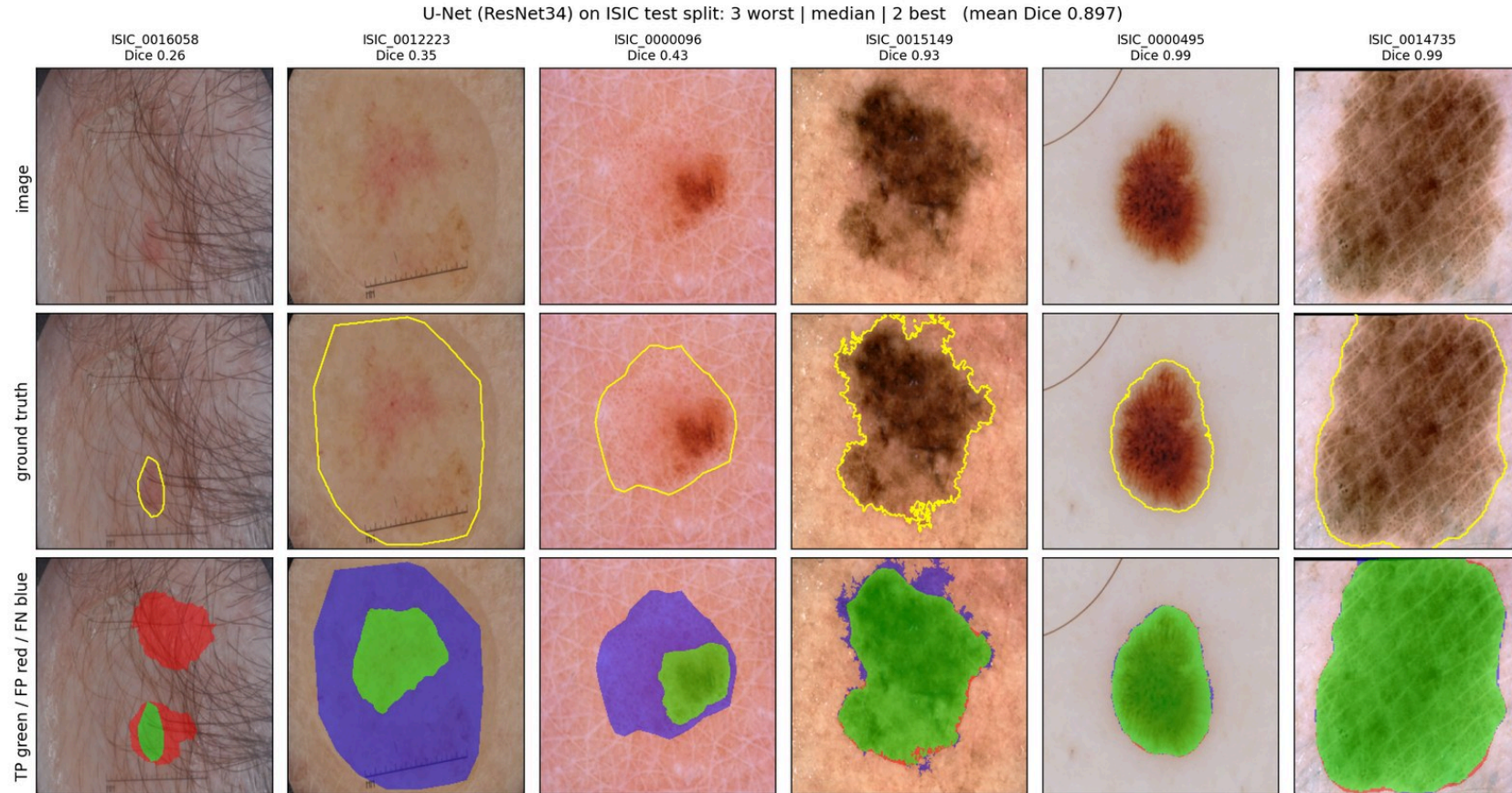
A good knife thrower misses on purpose. Our classifier misses the lungs by chance.

The placebo test for heat maps



Same images, same layer, same method. Top: trained. Bottom: all weights re-randomised. Mean map correlation -0.03 (per case $0.00 / -0.47 / 0.17 / 0.19$) • Adebayo et al. 2018

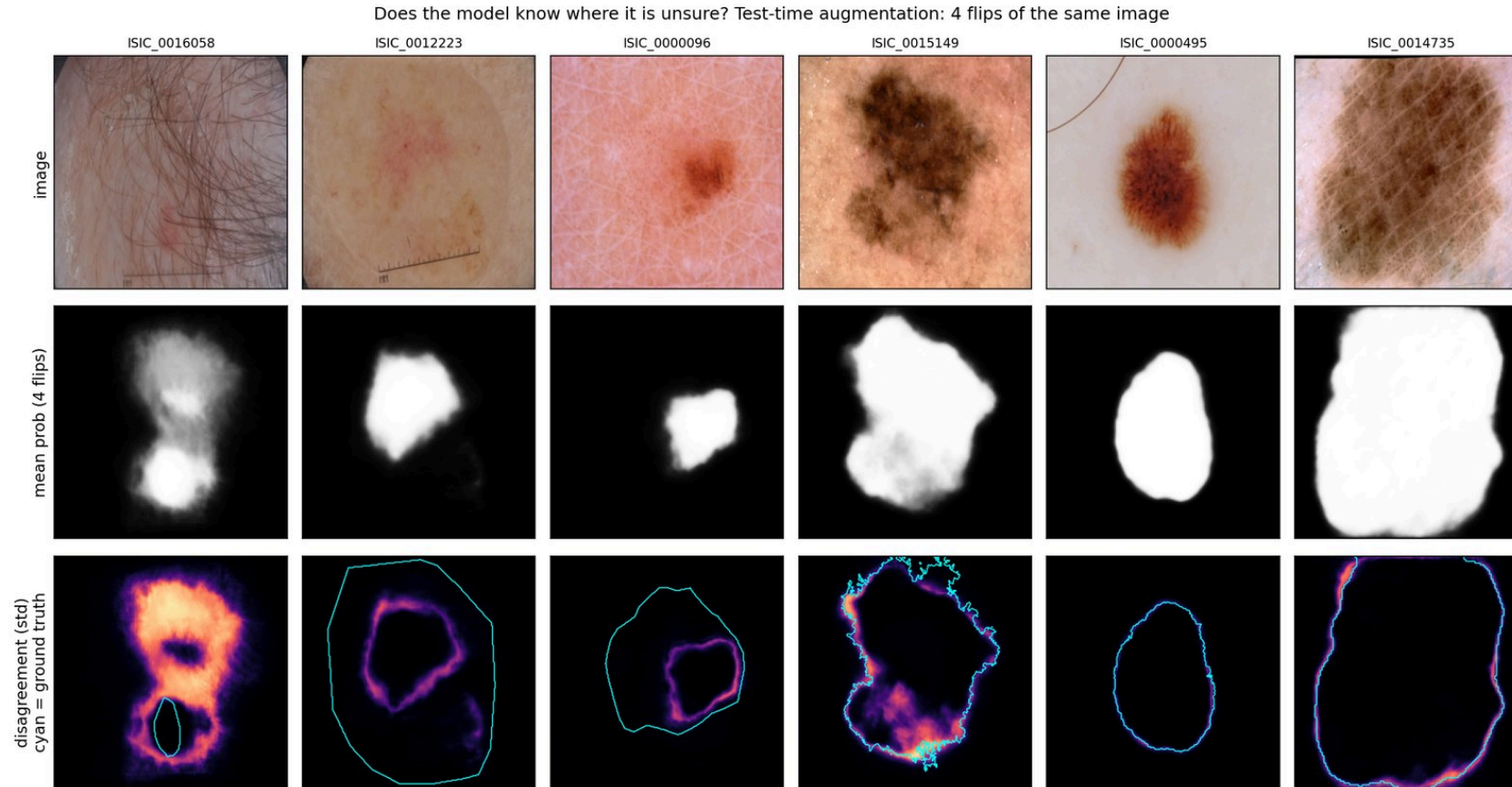
Where does it fail?



392 test images. The three worst, left: hair · the annotation outlines the whole glass plate · no contrast. Green correct, red too much, blue missed.

Mean Dice 0.897. The mean is where the 22 failures hide.

Does it know when it is wrong?



The same six images. Middle row: mean of four flipped passes. Bottom row: where the four disagree, bright = disagreement, cyan = reference outline. Across 392 images, disagreement vs Dice: Spearman $\rho = -0.36$.

Where the four answers differ, the errors sit. The hair blob glows. The wrong annotation does not.

Why does it say that?

Same image, same model, five prompts: what the VLM calls it
(green = matches histology, red = does not, grey = no label found)

ISIC_0002469 (histology: Benign)	Benign	Benign	Benign	Benign	Benign
ISIC_0014274 (histology: Benign)	Malignant	Malignant	Malignant	Malignant	Benign
ISIC_0012744 (histology: Benign)	Malignant	Benign	Malignant	Malignant	Benign
ISIC_0012455 (histology: Benign)	Malignant	Malignant	Malignant	Malignant	Malignant
ISIC_0013975 (histology: Malignant)	Malignant	Malignant	Malignant	Malignant	Benign
ISIC_0013618 (histology: Malignant)	Malignant	Malignant	Malignant	Malignant	Benign
ISIC_0014324 (histology: Malignant)	Malignant	Malignant	Malignant	Malignant	Benign
ISIC_0000043 (histology: Malignant)	Malignant	Malignant	Malignant	Malignant	Malignant
	naive	few_shot	chain_of_thought	structured_output	combined

Dermoscopic Examination Report

Patient ID: [Redacted]

Date of Examination: [Redacted]

Anatomic Site: [Not specified ...]

the "naive" prompt, verbatim · 6 of 8 reports look like this

Qwen3-VL · eight test images, four benign, four melanomas · only the instruction changes · green: matches histology

Same pixels, five prompts: 5 of 8 images change their label. The prompt decides, not the image.

Same pixels, different patient

16 of 16

answers: **malignant**.

Two melanomas, four patients each:
original · age 25 · age 75 · no metadata.

age 25: "Spitz nevus in young female is possible but cannot be ruled out clinically."

age 75: "High risk for melanoma given **age**, size, and dermoscopic atypia."

original: "asymmetry in **elderly** patient."

chain-of-thought reasoning for the same image, verbatim

The verdict never moved. The reasoning rewrote itself around it.

The colour of nothing

The colour of nothing: one flat square, same prompt and metadata as the melanoma, what the VLM calls it

	ISIC_0013975 (age 80, female, site Head and neck)	ISIC_0013618 (age 65, male, site Trunk)
grey (reference)	benign / reassure conf low	benign / reassure conf low
black	benign / reassure conf low	benign / reassure conf low
white	benign / reassure conf low	benign / reassure conf low
skin	benign / reassure conf low	benign / reassure conf low
light brown	suspicious / excision conf low	benign / reassure conf low
dark brown	benign / reassure conf low	benign / reassure conf low
red	suspicious / excision conf low	suspicious / excision conf low



```
"image_quality":  
"poor",  
"structures": [],  
"impression":  
"benign",  
"recommendation":  
"reassure"
```

the melanoma's prompt on a grey square

Seven flat squares, same prompt and metadata as the two melanomas. Every answer says "image quality poor", "confidence low".

Nothing to see: 10 of 14 times "benign, reassure". Red: surgery.

What we take home

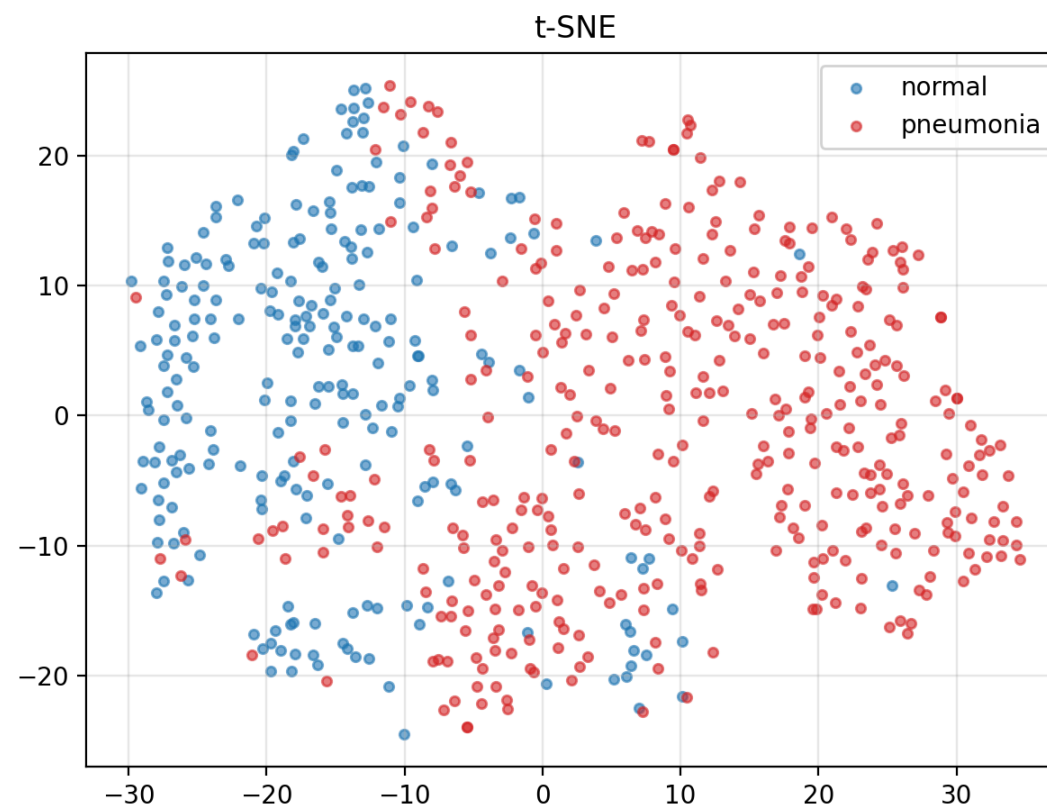
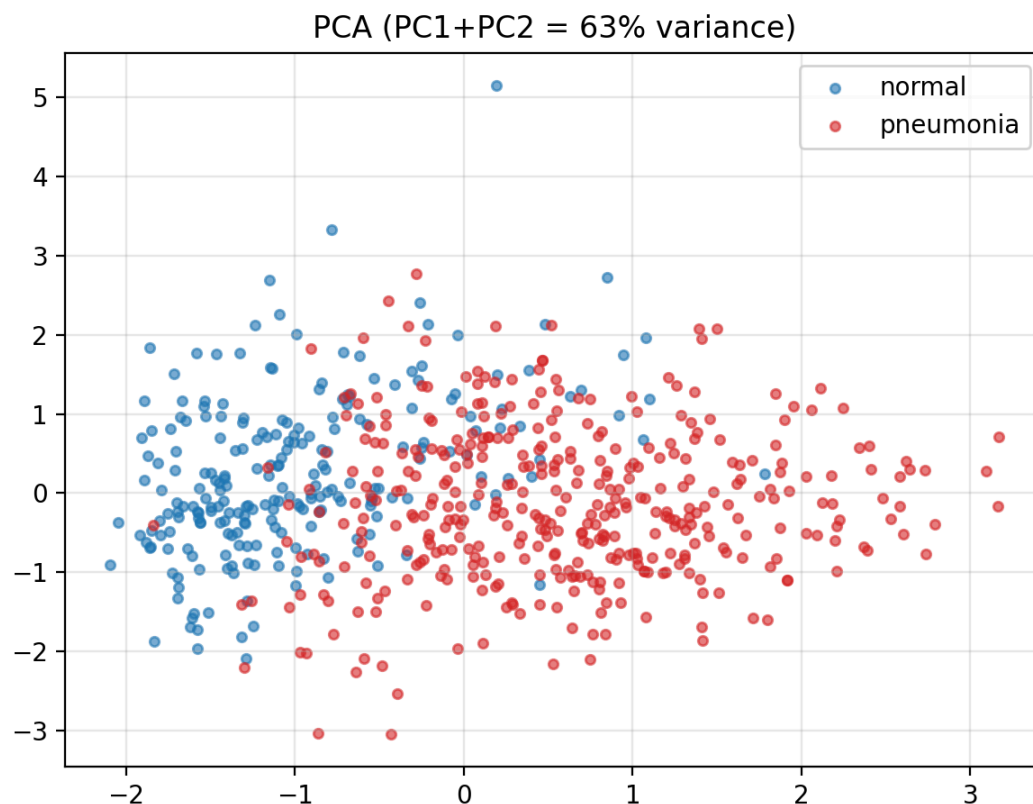
1. **A map is not a proof.** Run the control: random weights, a lung mask, another prompt, another patient, no image.
2. **For segmentation, measure errors** instead of admiring heat maps.
3. **A fluent reason is the most dangerous explanation of all.**

Appendix

Three more controls, for the discussion.

Does it separate the classes internally?

Features of layer2, test set



128 features of layer2 per image, squeezed to two dimensions. PCA: 63% of the variance in two axes. Overlap is where the errors live.

Same pixels, eight models

Same 8 images, same prompt (structured_output), different models: what each VLM calls it
(green = matches histology, red = does not, grey = no label found)

	ISIC_0002469 (Benign)	ISIC_0012455 (Benign)	ISIC_0012744 (Benign)	ISIC_0014274 (Benign)	ISIC_0000043 (Malignant)	ISIC_0013618 (Malignant)	ISIC_0013975 (Malignant)	ISIC_0014324 (Malignant)
Qwen3-VL 235B (Referenz) [5/8]	Benign	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant
Claude Sonnet 5 [4/8]	Benign	Malignant	Benign	Benign	Benign	Benign	Benign	Malignant
Claude Haiku 4.5 [5/8]	Benign	Malignant	Benign	Malignant	Malignant	Malignant	Benign	Malignant
Amazon Nova Pro [4/8]	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant
Mistral Pixtral Large [4/8]	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant	Malignant
Gemma 3 27B [5/8]	Malignant	Malignant	Benign	Benign	Malignant	Benign	Malignant	Malignant
GPT-6 Astra [5/8]	Benign	Benign	Malignant	Malignant	Benign	Malignant	Malignant	Malignant
Grok 4.6 [2/8]	Malignant	Malignant	Malignant	Malignant	Benign	Malignant	Malignant	Benign

ISIC_0002469, benign, age 15,
differential:

Haiku 4.5: "dermatofibroma"

Sonnet 5: "café-au-lait macule"

GPT-6: "Spitz nevus"

Gemma 3: "melanoma in situ"

Pixtral: "melanoma, Reed nevus"

same image, same prompt · five models, five
diseases

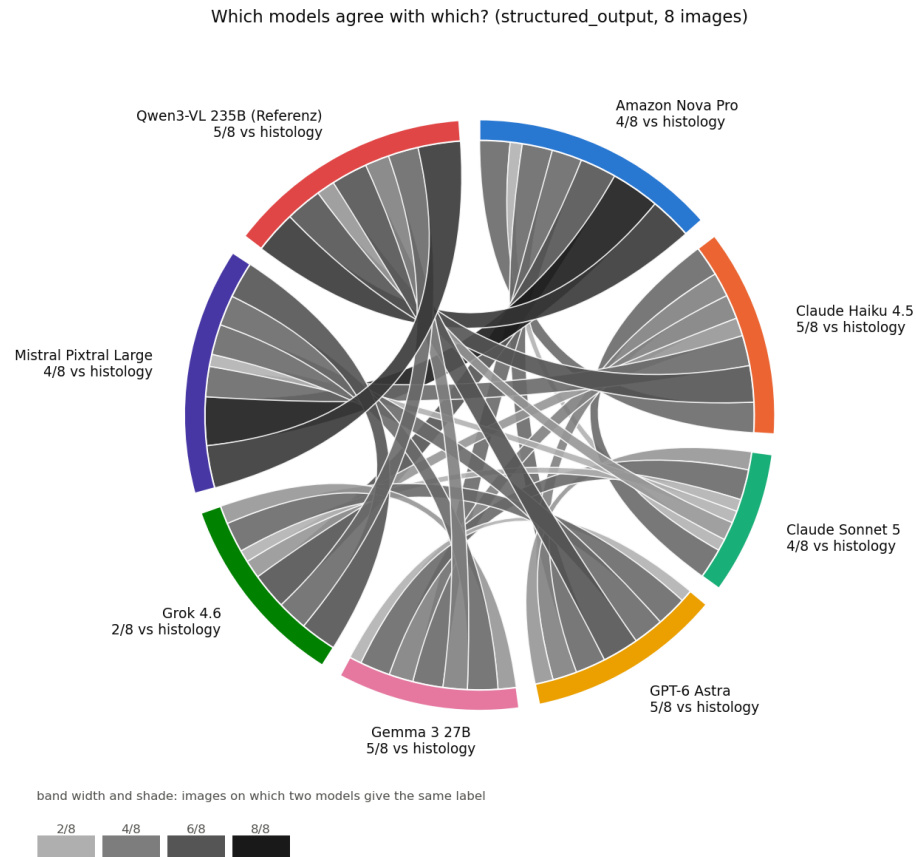
"impression": "benign",
"recommendation": "excision"

Sonnet 5 on ISIC_0000043, a melanoma:
harmless, but please cut it out

Same 8 images and the same structured prompt as before, only the model changes · best score **5 of 8**, no model beats Qwen · **8 of 8** images get contradictory labels across models · Nova, Pixtral and Grok call all four benign lesions malignant, Sonnet 5 calls three of four melanomas benign · majority vote: 5 of 8

Eight second opinions. Not one of them is a second opinion on the image; each is a first opinion on its own training data.

Who agrees with whom?



Band width: number of the 8 images on which two models give the same label. Arc length: how often a model agrees with the others at all.

Nova Pro · Pixtral: 8 of 8
 Qwen · Pixtral: 7 of 8
 Qwen · Nova Pro: 7 of 8

the three closest pairs are the three models that say "malignant" to almost everything

Sonnet 5 · Pixtral: 2 of 8
 Sonnet 5 · Nova Pro: 2 of 8

the loneliest model is the one that under-calls

run 2 → run 3, same settings:
 Sonnet 5 5/8 → 4/8 · Nova 5/8 → 4/8
 majority vote 6/8 → 5/8

we ran it twice · two models changed their score without anything changing

The models that agree most are the ones that say "malignant" to everything. Agreement is not accuracy.

